

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

10

HEAT-PART-3



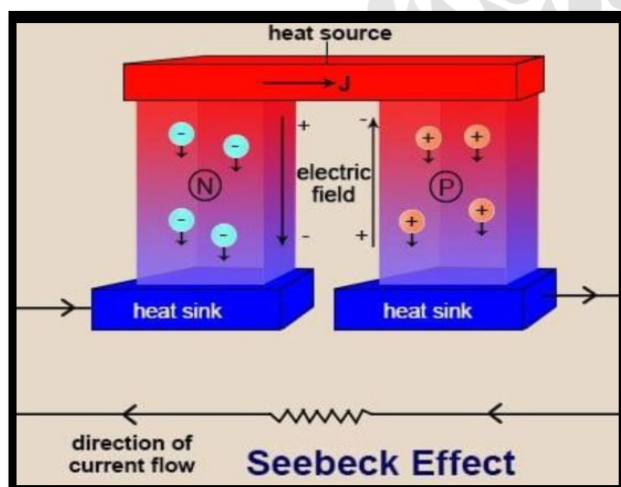
HEAT-PART-3

Thermal Expansion

Thermal Expansion is the tendency of a substance to change its dimensions (length, area, or volume) when its temperature changes, usually expanding when heated and contracting when cooled.

- Substances expand on heating and contract on cooling

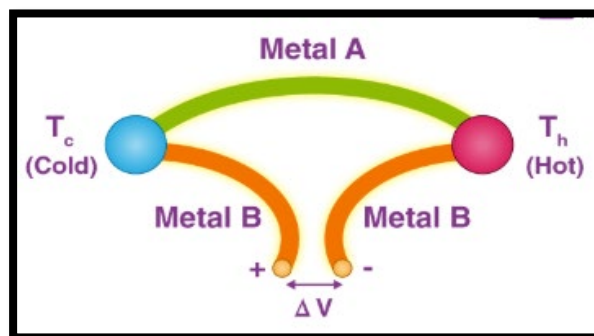
See Beck Effect (Thermoelectric Effect)



The Seebeck effect is the phenomenon in which an electric voltage (emf) is generated in a circuit made of two dissimilar conductors when their junctions are kept at different temperatures.

Basic Principle

- When two different metals are joined, they form two junctions:
 - Hot junction (higher temperature)
 - Cold junction (lower temperature)
- Due to temperature difference, charge carriers (electrons/holes) diffuse from hot to cold region.
- This creates a potential difference (voltage) in the circuit.



Types of thermal expansion

- Linear Expansion
- Areal Expansion
- Volume Expansion

I. Linear Expansion:

Change in length of a solid.

Example: Railway tracks expand in summer, so gaps are left between rails.

Coefficient of linear expansion α

$$\alpha = \frac{\Delta l}{l_{\text{original}} \times \Delta T}$$

II. Areal Expansion

Change in area of a surface.

Example: Metal plates expand when heated.

Coefficient of areal expansion (β)

$$\beta = \frac{\Delta A}{A_{\text{original}} \times \Delta T}$$

III. Volume Expansion

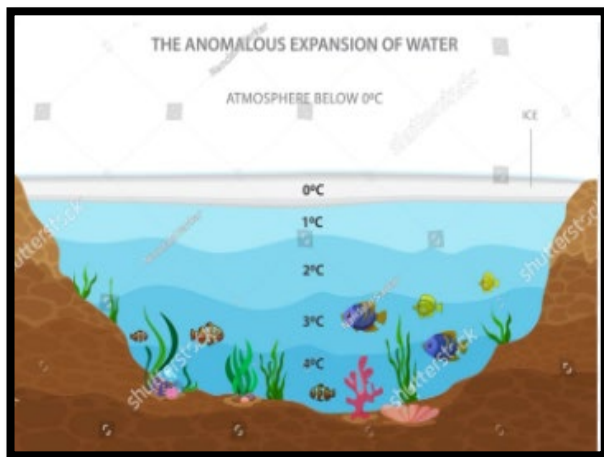
Change in volume of a substance (solids, liquids, gases).

Example: Liquids expand in a thermometer when heated.

Coefficient of volume expansion (γ)

$$\gamma = \frac{\Delta V}{V_{\text{original}} \times \Delta T}$$





Applications for thermal Expansion in Solids

1. Expansion Joints in Structures

- Bridges, flyovers, and buildings are provided with expansion joints.
- These joints allow materials to expand and contract with temperature changes.
- Prevents cracking, bending, or structural failure.
- Especially important in regions with extreme temperature variation.

2. Railway Tracks and Roads

- Small gaps are left between railway tracks.
- In hot weather, rails expand; gaps prevent buckling or warping.
- Similarly, concrete roads are constructed with expansion gaps.

3. Thermometers (Liquid Expansion)

- Devices like mercury or alcohol thermometers work on volume expansion of liquids.
- When temperature rises, the liquid expands and moves up the capillary tube.
- Used in medical, laboratory, and environmental measurements.

4. Bimetallic Strip (Temperature Control Devices)

- Made of two different metals with different expansion rates.
- On heating, one expands more, causing bending.
- Used in:
 - Electric irons
 - Thermostats
 - Circuit breakers
- Helps in automatic temperature regulation.

5. Loosening Tight Lids and Mechanical Fittings

- Metal lids expand when heated, making them easier to open.
- Used in workshops for fitting/removing metal parts.
- Example: Heating a nut makes it expand and loosen from a bolt.

6. Overhead Electric Wires

- Wires are kept slightly loose during installation.
- In summer, they expand; in winter, they contract.
- Prevents snapping due to contraction in cold weather.

7. Riveting Process in Engineering

- Rivets are heated before joining metal plates.
- On cooling, they contract and form a tight, strong joint.
- Widely used in bridges, boilers, and shipbuilding.

8. Glass and Metal Fitting (Thermal Stress Use)

- Metal rims are heated to fit tightly around wooden wheels or glass objects.
- After cooling, they contract and hold the object firmly.

9. Expansion in Pipelines and Machinery

- Oil pipelines and steam pipes are designed with loops or bends.
- This allows expansion and prevents damage due to thermal stress.

10. Hot Air Balloons (Gas Expansion)

- Air expands when heated, becoming less dense.
- This provides lift in hot air balloons.
- Demonstrates practical use of thermal expansion of gases.

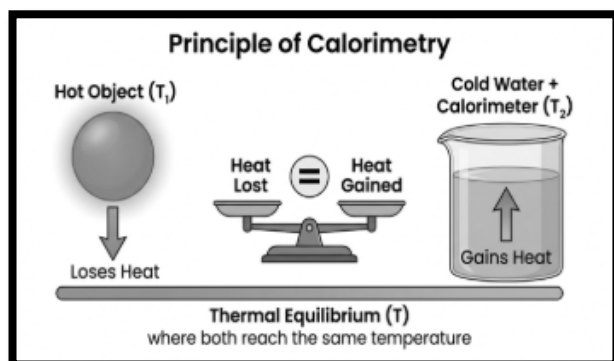
Effect of temperature on the time period of a simple pendulum:

- A pendulum clock keeps proper time at temperature q .
- If temperature is increased to $q' (> q)$ then due to linear expansion, length of pendulum increases and hence its time period will increase.



LAW OF MIXTURES: Principle of calorimetry

- When two bodies at different temperatures are mixed, heat will be transferred from body at higher temperature to a body at lower temperature till both acquire same temperature.
- The body at higher temperature releases heat while body at lower temperature absorbs it, so that (Principle of calorimetry) Principle of calorimetry represents the law of conservation of heat energy.



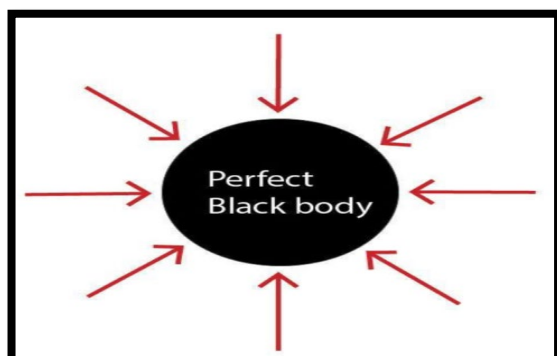
$E \propto T^4$, E = energy emitted, T = Temperature

Ideal Black Body

- A body surface which absorbs all incident thermal radiation at low temperature, irrespective of their wavelength and emits out all these absorbed radiations at high temperature is assumed to be an ideal black body surface.

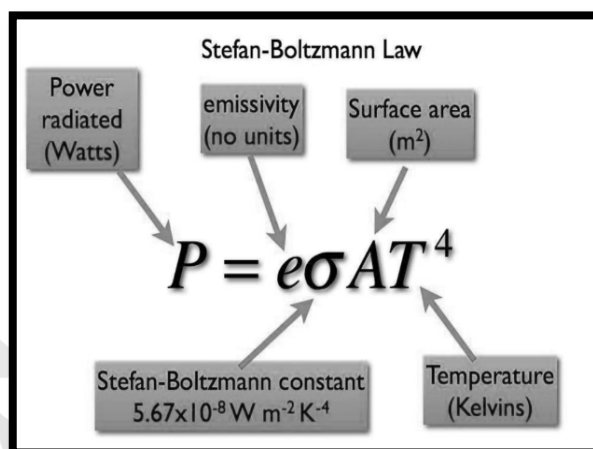
Key Characteristics

- Perfect absorber $\rightarrow$ absorbs 100% radiation (no reflection or transmission)
- Perfect emitter $\rightarrow$ emits maximum radiation
- Emissivity ($e = 1$)
- Radiation depends only on temperature, not on material
- Appears perfectly black at all temperatures



STEFAN'S LAW

- The amount of radiation emitted per second per unit area by ideal black body is directly proportional to the fourth power of its absolute temperature.



WEIN'S DISPLACEMENT LAW

- The wavelength corresponding to maximum emission of radiation decrease with increasing temperature $\left[\lambda_m \propto \frac{1}{T}\right]$
- This is known as Wein's displacement law. $\lambda_m T = b$ where b is Wein's constant $= 2.89 \times 10^{-3} \text{ mK}$

Dimensions of b : $M^0 L^1 T^0 \theta^1$

NEWTON'S LAW OF COOLING

- Rate of cooling $\left(\frac{dT}{dt}\right)$ is directly proportional to excess of temperature of the body over that of surrounding. $-\left[\text{When } (T - T_0) \text{ is } 35^\circ\text{C}\right] - \frac{dT}{dt} \propto (T - T_0)$
- T = temperature of body [all temperatures in $^\circ\text{C}$]
- T_0 = temperature of surrounding.
- $T - T_0$ = excess of temperature ($T > T_0$)
- If the temperature of body is decreased by dT in time dt then rate of fall of temperature $-\frac{dT}{dt} = K(T - T_0)$
- Where negative sign indicates that the rate of cooling is decreasing with time.

Law of Thermodynamics

The Laws of Thermodynamics are fundamental principles in Thermodynamics that govern how energy is transferred and transformed in physical systems. They are essential for physics, chemistry, engineering, and even environmental science.

I. Zeroth Law of Thermodynamics

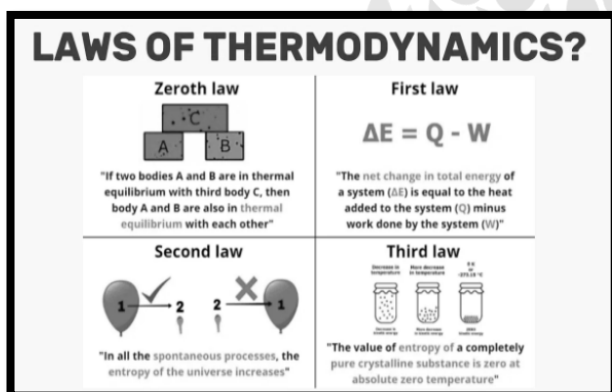
- If two systems are each in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.

Meaning:

- It defines the concept of temperature.
- Basis for using thermometers.

Example:

- If body A = body B (same temperature) and body B = body C
- Then A = C



Key Concept: Entropy (S)

- Entropy measures disorder/randomness.
- In an isolated system, entropy always increases.

Meaning:

- Natural processes are irreversible.
- Explains why efficiency is always less than 100%.

Example:

- Heat flows from hot tea to surrounding air, not the reverse.

IV. Third Law of Thermodynamics

- As temperature approaches absolute zero, the entropy of a perfect crystal approaches zero.

Meaning:

- Absolute zero (0 Kelvin) is unattainable.
- System becomes perfectly ordered.

II. First Law of Thermodynamics (Law of Energy Conservation)

- Energy can neither be created nor destroyed; it can only change from one form to another.

Where:

- Heat supplied
- Internal energy
- Work done

Meaning:

- Heat given to a system is used to increase internal energy or to do work.

Example:

- Steam engine converts heat into mechanical work.
- Human body converts food into energy.

III. Second Law of Thermodynamics

- It is impossible to convert all heat into work without any loss.
- Statement (Clausius form):
- Heat cannot flow spontaneously from a colder body to a hotter body.

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